

HCR Project

How to Write a Research Paper

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Project 09 - Junior Research Mentorship Programme
The Hong Kong Polytechnic University (PolyU)

Projects

All project information is released on the Google Drive.

Project 1: Privacy-Preserving Smart Classroom System Using Sensors and MQTT for Adaptive Learning Spaces

Project 2: Intelligent Practical Assessment System for Advanced Physics or Chemistry Laboratory Instruments

TL

Week 1

We will cover all basic concepts and tools for building an AloT system, including MQTT, sensors, microcontrollers, and programming.

A worksheet should be submitted before 5:59 AM on the next Monday.

Week 2

We will continue with more concepts and starting to build simple demos.

A worksheet should be submitted before 5:59 AM on the next Monday.

You should start thinking about team formation and project selection.

TL

Week 3 (We are here)

We will divide groups and start building. Each group will choose a project theme, design the system architecture, and implement a working prototype by the end of July.

Each group should have 3-4 members

You should make your final group decision and submit the group member list before 5:59 AM on the next Monday.

Week 4

Finalize the system architecture, define the interface between modules, and assign responsibilities to each team member.

A final architecture diagram and work allocation plan should be submitted before 5:59 AM on the next Monday.



Structure of Paper

Usual Structure of an Essay / Paper

- Abstract
- Introduction
- Literature Review
- Methodology
- Results
- Discussion
- Conclusion

Abstract

Appears at the beginning of the paper, but is usually written at the end.

It is a concise summary of the entire paper, including the research question, methods, results, and conclusions.

The reader will decide whether to read your paper or not based on the abstract, so it should be short, clear, informative, and engaging.

Introduction

The introduction explains the background of the topic and why the problem is important.

It should clearly state the main research problem, the objective of the work, and the overall direction of the paper.

After reading the introduction, the reader should understand what the paper is about and why it is worth studying.

Literature Review

The literature review summarizes related studies, existing systems, or previous methods.

Its purpose is to show what has already been done and what limitations still remain in the field.

This section helps justify why your project is needed and how it is different from earlier work.

Methodology

The methodology explains how the project or study is carried out.

It usually includes the system design, tools, sensors, software, data collection process, and evaluation method.

This section should be clear enough that another person can understand how the work was implemented.

Results

The results section presents the main outcomes of the work.

This may include measurements, observations, tables, screenshots, or performance comparisons.

The focus here is on reporting what happened, rather than explaining in detail why it happened.

Discussion

The discussion interprets the results and explains their meaning.

In this section, you can describe whether the project met its goals, what worked well, what problems appeared, and what limitations remain.

This is where you connect your results back to the research question and show your understanding of the project.

Conclusion

The conclusion gives a short final summary of the whole paper.

It should restate the main problem, briefly mention the method and findings, and explain the overall contribution of the work.

A good conclusion may also suggest possible future improvements or next steps for the project.